

Feiyu Lu, PhD

RESEARCH SCIENTIST

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About Me

Feiyu Lu is a Research Scientist in the Immersive Technology & Spatial Computing Team, Global Technology Applied Research Group at JPMorganChase. He obtained his Ph.D. degree at Virginia Tech advised by Prof. Doug A. Bowman. His research interest lies in the intersections of **Spatial/Multimodal Interaction**, **Human-AI Interaction**, **Artificial Intelligence / Machine Learning**, and **Augmented/Virtual/Mixed Reality**.

Education

Virginia Polytechnic Institute and State University (Virginia Tech)

Blacksburg, VA, USA

Ph.D. in Computer Science & Applications

Aug 2018 - May 2023

- Research Group: **3D Interaction Group**
- Advisor: **Prof. Doug A. Bowman**
- Dissertation: 📄 **Glanceable AR - Towards a Pervasive and Always-On Augmented Reality Future**
- Committee Members: Prof. Joseph L. Gabbard, Prof. Wallace Lages, Prof. Sang Won Lee, Prof. Yalong Yang, Prof. Steven K. Feiner

Xi'an Jiaotong-Liverpool University (XJTLU)

Suzhou, Jiangsu, China

B.Eng in Computer Engineering

Sep 2014 - Jun 2018

- Advisor: **Prof. Hai-Ning Liang**
- Graduated with First-Class Honor (Top 5%)

Research Experience

Immersive Technology & Spatial Computing Team, JPMorganChase

New York, NY, USA

Research Scientist

May 2023 - Current

- I am working as an Applied Research Scientist in the Global Technology Applied Research Group at JPMC.
- I investigate novel integrations of immersive technology, interactive wall displays, and AI/ML techniques such as reinforcement learning, large language models, and large multimodal models to addressing concrete business problems and advancing our clients & employee experiences.
- I work with internal stakeholders, build novel intelligent systems, conduct user-centric research, and deliver tech-transfer-ready proof-of-concept systems to boost employee productivity and client engagement.
- I have contributed to six patents, five full papers, and three extended abstracts at top conferences.
- Manager: **Blair MacIntyre, Managing Director**

Reality Labs Research, Meta Inc.

Redmond, WA, USA

Research Scientist Intern

May 2022 - Aug 2022

- I worked with teams at Reality Labs Research to perform user experience research on spatial interactions.
- I researched how user and system factors may influence the perceived human-AI interaction experience in immersive spatial environments.
- I conducted a remote study with over 80 Meta Quest consumers with deep insights generated from stratified sampling, mixed-method research, and generalized linear mixed modeling.
- The work was published and presented at ISMAR'23, a top conference in the field of computer graphics and usability.

Reality Labs Research, Meta Inc.

Remote

Research Intern

May 2021 - Aug 2021

- I worked with teams at Reality Labs Research to perform usability evaluations on intelligent user interfaces.
- I conducted workshops with UX researchers and designers to identify pain-points during spatial multi-modal interactions.
- I conducted a study with over 40 everyday Meta Quest consumers to distill usability insights.
- The work was patented and published at CHI'22, a top conference in the field of human-centric computing.

3D Interaction Group

Blacksburg, VA, USA

Graduate Researcher

Aug 2018 - May 2023

- I worked as a GRA in the 3D Interaction Group directed by Prof. Doug A. Bowman.
- I worked on exploring interaction techniques for lightweight activation of everyday AR content.
- I worked on designing, implementing and evaluating general-purpose systems for AR HWDs.
- I worked on developing efficient and unobtrusive information access strategies for AR HWDs.
- I worked on evaluating gaze direction visualization techniques in wide-area collaborative AR environments.
- Advisor: Prof. Doug A. Bowman

Selected Publications

I published the majority of my work in premiere conferences / journals on spatial user interactions / human-computer interactions / immersive technologies. The typical acceptance rates of these venues are around 25%. Below is a selected list of my publications.

S-TIER: Situated-Traceable Insights in Extended Reality for Hybrid Crisis Management

Feiyu Lu*, Cheng Yao Wang*, Leonardo Pavanatto, Fannie Liu, David Saffo, Benjamin Lee, Mengyu Chen, Blair MacIntyre

“Where Did My Apps Go?” Supporting Scalable and Transition-Aware Access to Everyday Applications in Head-Worn Augmented Reality

Feiyu Lu, Leonardo Pavanatto, Shakiba Davari, Lei Zhang, Lee Lisle, Doug A. Bowman

IEEE Transactions on Visualization and Computer Graphics. 2025

DOI: 10.1109/TVCG.2024.3493115

Adaptive 3D UI Placement in Mixed Reality Using Deep Reinforcement Learning

Feiyu Lu*, Mengyu Chen*, Hsiang Hsu, Pranav Deshpande, Cheng Yao Wang, Blair MacIntyre

Extended Abstracts of the CHI Conference on Human Factors in Computing Systems, 2024, Honolulu, HI, USA

DOI: 10.1145/3613905.3651059

In-the-Wild Experiences with an Interactive Glanceable AR System for Everyday Use

Feiyu Lu, Leonardo Pavanatto, Doug A. Bowman

Proceedings of the 2023 ACM Symposium on Spatial User Interaction (SUI), 2023, Sydney, NSW, Australia

DOI: 10.1145/3607822.3614515

Exploring the Impact of User and System Factors on Human-AI Interactions in Head-Worn Displays

Feiyu Lu, Yan Xu, Xuhai Xu, Brennan Jones, Laird Malamed

2023 IEEE International Symposium on Mixed and Augmented Reality (ISMAR), 2023, Sydney, NSW, Australia

DOI: 10.1109/ISMAR59233.2023.00025

Exploring Spatial UI Transition Mechanisms with Head-Worn Augmented Reality

Feiyu Lu, Yan Xu

Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (CHI), 2022, New Orleans, LA, USA

DOI: 10.1145/3491102.3517723

Evaluating the Potential of Glanceable AR Interfaces for Authentic Everyday Uses

Feiyu Lu, Doug A. Bowman

2021 IEEE Virtual Reality and 3D User Interfaces (VR), 2021, Lisbon, Portugal

DOI: 10.1109/VR50410.2021.00104

Exploration of Techniques for Rapid Activation of Glanceable Information in Head-Worn Augmented Reality

Feiyu Lu, Shakiba Davari, Doug A. Bowman

Proceedings of the 2021 ACM Symposium on Spatial User Interaction (SUI), 2021, Virtual Event, USA

DOI: 10.1145/3485279.3485286

Glanceable AR: Evaluating Information Access Methods for Head-Worn Augmented Reality

Feiyu Lu, Shakiba Davari, Lee Lisle, Yuan Li, Doug A. Bowman

2020 IEEE Conference on Virtual Reality and 3D User Interfaces (VR), 2020, Atlanta, GA, USA

DOI: 10.1109/VR46266.2020.00113

Awards

2025	VR Dissertation Award , IEEE Visualization and Graphics Technical Community (VGTC)	Saint-Malo, France
2024	Prolific Inventor , Global Technology Applied Research	JPMorganChase
2023	Best Paper Honorable Mention Award (Top 5%) , CHI'23	Hamburg, Germany
2021	Best Paper Honorable Mention Award (Top 5%) , SUI'21	Virtual Event
2021	Finalist (Top 3.5% of 2,163 applicants) , Facebook Ph.D. Fellowship	Virtual Event

Patents

2025	System and Method for Long-Term Video Analysis and Context-Aware Search Using Vision-Language Models , No. 19/222,406	with JPMorganChase
2024	Privacy-preserving Methods for Remote Space Sharing in Enterprise Environment , No. 18/774,705	with JPMorganChase
2024	Systems and Methods for Hybrid Cross-Reality Collaboration Around an Large Interactive Display , No. 18/882,495	with JPMorganChase
2024	HybridPortal: Enabling Hybrid Social Interactions for Hybrid Events , No. 18/882,477	with JPMorganChase
2024	Systems and Methods for Audience Feedback Guided Mixed Reality , No. 18/769,647	with JPMorganChase
2023	Systems and Methods for Placement of Virtual Content Objects in an Extended Reality Environment Based on Reinforcement Learning , No. 18/407,003	with JPMorganChase
2021	Dynamic Widget Placement Within an Artificial Reality Display , No. 17/747,767	with Meta Inc.
2020	Interacting with Glanceable Information in Wearable Augmented Reality , No. 63/147,805	with Virginia Tech

Skills

Research	3D Interaction, Human-AI Interaction, Multimodal Input, Spatial Computing, Extended Reality, Human-Computer Interaction.
Programming/Platform	Python, C#, JavaScript/TypeScript, Swift, WebXR, VisionOS, Meta Quest.
Tools	Unity3D, BabylonJS, Arduino, Blender, R, SPSS, Adobe Suites.
Method	Rapid Prototyping, Mixed-methods Research, Inferential Statistical, Thematic Analysis, Experimental Design, Usability Evaluation, Performance Modeling.